

**Minor Project Synopsis Report**

**NeuroTrade**

**Project Category: Research Based**

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## 1. ABSTRACT

The stock market is a very volatile system where the prices of assets are driven by a set of complex interactions between quantitative elements (historical price patterns) and qualitative elements (news, market sentiment, and global events). Retail investors find it very difficult to make informed decisions due to the dispersed nature of these sources of information—technical analysis is available on one platform, while financial news is available on other platforms. The objective of this project, AI-Driven Stock Market Sentiment & Prediction System, is to fill this gap by providing a comprehensive web dashboard.

We use Long Short-Term Memory (LSTM) networks, which are a type of Recurrent Neural Network (RNN), to predict short-term price patterns based on historical data. At the same time, we use FinBERT, a pre-trained Natural Language Processing (NLP) model, to analyze the headlines of financial news articles and provide a real-time sentiment score (Positive/Negative). By integrating these pieces of information into a single "Prediction & Sentiment" dashboard, the system enables users to view market trends from a holistic perspective. The project is developed using the FARM Stack (FastAPI, React, MongoDB) to make it scalable and capable of providing real-time outputs.

**KEYWORDS:** Stock Prediction, Sentiment Analysis, LSTM, FinBert, NLP, Financial Modeling.

## 2. INTRODUCTION

The financial markets have witnessed a dramatic shift with the introduction of High Frequency Trading and Algorithmic Trading. However, for the common retail investor or student, the analysis tools that are available are still largely manual. The conventional analysis techniques can be categorized into two schools of thought: Technical Analysis, which focuses on charts and patterns, and Fundamental Analysis, which focuses on news and announcements related to the company.

In recent years, "Sentiment Analysis" has become the third essential pillar. A news headline or a tweet can bring down a stock price in minutes, often contrary to technical analysis. This project proposes a system that uses Artificial Intelligence to automatically analyze both the price history and news sentiment. By analyzing an enormous amount of data—historical stock prices and thousands of news articles—our system will give a concise, actionable piece of information that would take humans hours to compile.

### **3. MOTIVATION**

In the post-pandemic world, there has been a huge rise in retail participation in the stock market (for example, through apps such as Zerodha/Robinhood). Unfortunately, research indicates that more than 80% of retail participants end up losing money. The main cause of this is Information Overload.

Investors cannot possibly read every news article that is published about their company.

They do not have the mathematical skills to carry out time-series forecasting.

Current professional software (such as Bloomberg Terminals) is too expensive for students and individuals.

This is why there is a need for a free, accessible, and AI-driven software that can democratize access to high-end financial analysis. We wanted to create a system that not only displays information but also understands it, telling the user not only what the price is but also how the market feels about it.

## 4. LITERATURE REVIEW

### 1. Stock Price Prediction Using LSTM:

Moghar, A., & Hamiche, M. (2020) showed that LSTM networks perform better than conventional methods such as ARIMA in stock price prediction because LSTMs are capable of maintaining "memory" of long-term trends, making them highly suitable for time-series data.

Findings: LSTMs are successful in identifying non-linear patterns in stock prices but tend to fail when sudden external occurrences (news) happen.

### 2. Financial Sentiment Analysis (FinBERT):

Araci, D. (2019) proposed FinBERT, a BERT model pre-trained on a large financial dataset.

Findings: Conventional NLP models (such as the basic BERT model) tend to misunderstand financial terminology (for example, "losses narrowed" is positive in finance but negative in general English). FinBERT overcomes this issue by recognizing the financial context, achieving higher accuracy in classification.

### 3. Hybrid Approaches:

Recent research indicates that a combination of quantitative models (LSTM) and qualitative models (NLP) provides higher accuracy than using either model alone. Our project is based on this hybrid approach.

## 5. GAP ANALYSIS

| Existing Solutions                                 | Limitations (The Gap)                                                           | Our Proposed Solution                                                                                |
|----------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Traditional Stock Apps</b> (e.g., MoneyControl) | Show charts and news separately. User must mentally connect the dots.           | <b>Unified Dashboard:</b> Overlays sentiment scores directly onto price predictions.                 |
| <b>Generic Prediction Models</b>                   | Use simple Linear Regression or Moving Averages. Low accuracy on volatile data. | <b>Deep Learning (LSTM):</b> Uses advanced memory networks to detect complex time-based patterns.    |
| <b>Standard Sentiment Tools</b>                    | Use generic NLP (e.g., NLTK/VADER) which fails to understand financial context. | <b>Domain-Specific NLP (FinBERT):</b> Specifically trained to understand financial language nuances. |
| <b>Professional Terminals</b>                      | Extremely expensive (\$20,000/year). Inaccessible to students.                  | <b>Free &amp; Open Source:</b> Built on accessible web technologies (MERN/FastAPI).                  |

## 6. PROBLEM STATEMENT

There is no integrated and automated system available for retail investors and market enthusiasts to analyze stock market trends. They have to depend on various sources, such as one website for graphics and another for news, which causes delays in decision-making. Moreover, analyzing thousands of news headlines on a daily basis is not possible for a human. There is a great need for an automated system that can forecast stock price trends based on history and measure market sentiment based on news at the same time, and display the results in a simple and understandable graphical format.

: **Fragmentation of Financial Data:** Today, retail investors are forced to use various different platforms to collect information—using one website for technical analysis (e.g. TradingView) and another for financial news (e.g. MoneyControl). This makes analysis inefficient and causes delays in decision-making.

: **Information Overload:** The amount of financial news, tweets, and earnings releases on a daily basis is simply too much for a human to manually process. Investors are missing out on key news alerts that can affect stock prices.

: **Lack of Unified Analysis:** There is no free, readily available resource that integrates Quantitative Analysis (historical stock price analysis) with Qualitative Analysis (news sentiment analysis) into one interface. Currently, available resources either do not have the capability of deep learning or are too costly for students and retail investors.

: **Complexity of Technical Analysis:** New investors find it difficult to analyze complex technical analysis tools. They require a simplified "Prediction Score" based on AI to confirm their investment decisions



## 7. OBJECTIVES

: To Develop a Unified Web Dashboard: Design and build a responsive full-stack web application using the FARM Stack (FastAPI, React, MongoDB) that serves as a single point of access for both stock charts and news analytics.

: To Implement Domain-Specific Sentiment Analysis: Deploy FinBERT, a specialized Natural Language Processing (NLP) model, to accurately classify financial news headlines into Positive, Negative, or Neutral sentiments, overcoming the limitations of generic sentiment tools.

: To Forecast Short-Term Price Trends: Train and deploy a Long Short-Term Memory (LSTM) neural network on 5 years of historical OHLC data to predict stock price movements for the next 7 days.

: To Visualize Complex Financial Insights: Create interactive, real-time data visualizations (Candlestick charts, Sentiment Gauges) that allow users to instantly grasp market trends without needing deep financial expertise.

: To Automate Real-Time Data Aggregation: Build a robust backend pipeline that automatically fetches, processes, and updates stock prices and news feeds in real-time, ensuring users always have the latest data.

## 8. TOOLS & TECHNOLOGY USED

### 1. Programming Language: Python & JavaScript

Python (Backend/AI): Utilized for its rich libraries in Data Science (Pandas, NumPy) and Deep Learning (TensorFlow, PyTorch). It is the industry standard for financial modeling.

JavaScript (Frontend): Utilized through React.js for developing a dynamic and responsive user interface.

### 2. Frameworks & Libraries:

FastAPI: A modern, high-performance web framework for developing APIs with Python. It enables the use of asynchronous processing, which is essential for executing AI models without halting the server.

TensorFlow / Keras: Utilized for developing and training the LSTM neural network.

Hugging Face Transformers: Utilized for developing the pre-trained FinBERT model.

React.js: Utilized for the frontend dashboard to display interactive charts (via Recharts).

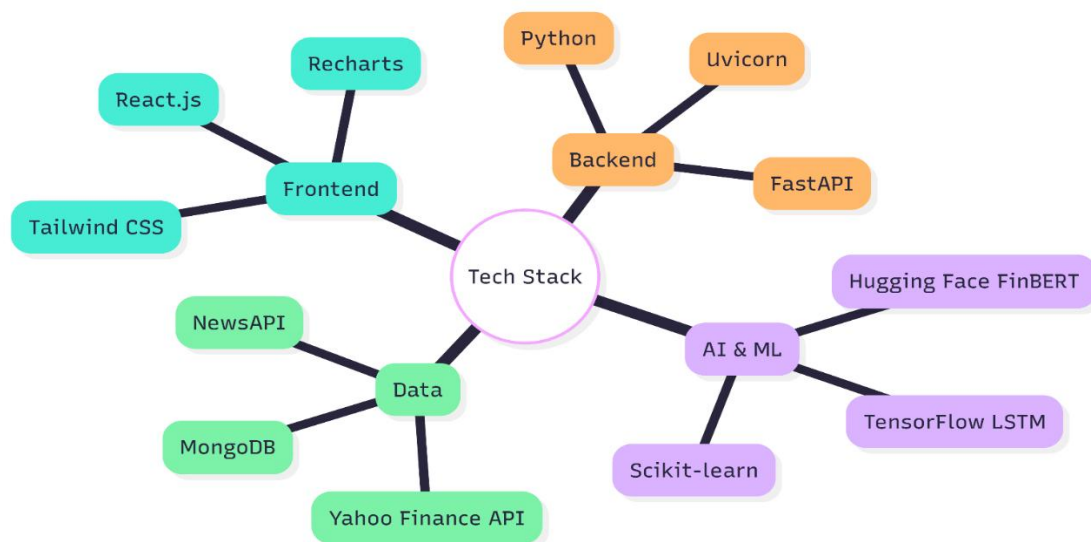
### 3. Data Sources:

yfinance API: Utilized for retrieving historical Open-High-Low-Close (OHLC) data.

NewsAPI: Utilized for retrieving real-time financial news headlines.

### 4. Database:

MongoDB: A NoSQL database for storing user search history and caching API responses to enhance load times.



## 9. METHODOLOGY

### 1. Quantitative Analysis (Price Prediction)

**Model:** Employing a Long Short-Term Memory (LSTM) network due to its capacity to handle long-term dependencies in time-series data.

**Process:** Feeding past OHLC data (5 years) from yfinance, Min-Max scaling (0-1) normalization, and 60-day chunks for predicting the closing price of the next day.

### 2. Qualitative Analysis (Sentiment Gauge)

**Model:** Applying FinBERT, a BERT transformer pre-trained on financial texts.

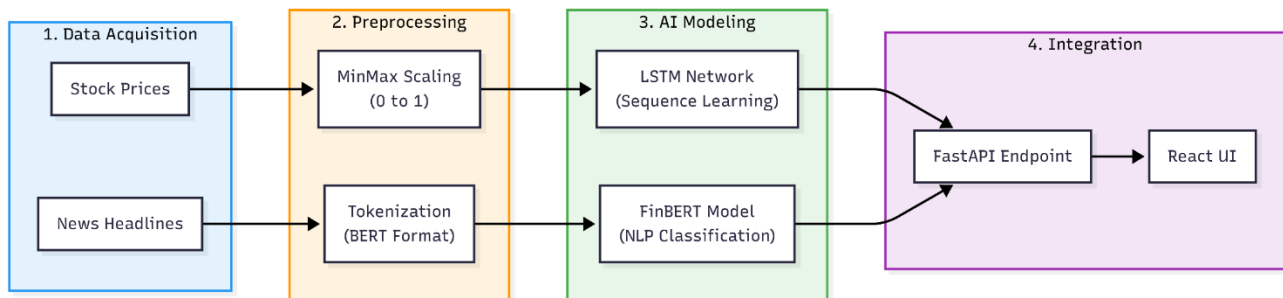
**Process:** Fetching real-time news headlines from NewsAPI, tokenization, and categorizing into Positive, Negative, or Neutral sentiments to obtain a weighted market sentiment score.

### 3. System Integration

**Backend:** Using a FastAPI server to run both AI models in parallel for low latency (<2 seconds).

**Frontend:** Employing a React.js dashboard to combine the numerical forecast and sentiment score, displaying them through interactive charts for comprehensive market analysis.

**Figure : Architectural Diagram**



## 10. REFERENCES

1. : **Araci, D. (2019).** *FinBERT: Financial Sentiment Analysis with Pre-trained Language Models*. arXiv preprint arXiv:1908.10063.

*Relevance:* This is the foundational paper for the FinBERT model you are using for sentiment analysis.

2. : **Moghar, A., & Hamiche, M. (2020).** *Stock Market Prediction Using LSTM Recurrent Neural Network*. *Procedia Computer Science*, 170, 1168-1173. <https://doi.org/10.1016/j.procs.2020.03.049>

*Relevance:* Validates the use of LSTM over standard algorithms for stock forecasting.

3. : **Fischer, T., & Krauss, C. (2018).** *Deep learning with long short-term memory networks for financial market predictions*. *European Journal of Operational Research*, 270(2), 654-669.

*Relevance:* Provides statistical proof that LSTMs are effective for financial time-series data.

4. : **Hugging Face.** (n.d.). *Transformers Documentation*. Retrieved January 2026, from <https://huggingface.co/docs/transformers/>

*Relevance:* Official documentation for the library used to implement the NLP model.

5. : **Ramírez, S.** (n.d.). *FastAPI Documentation*. Retrieved January 2026, from <https://fastapi.tiangolo.com/>

*Relevance:* Official documentation for the backend framework used in the project.

6. : **Abadi, M., et al. (2016).** *TensorFlow: A System for Large-Scale Machine Learning*. OSDI '16 Proceedings of the 12th USENIX conference on Operating Systems Design and Implementation.

*Relevance:* Cites the core Deep Learning framework used for the project.